

FEATURE REPORT

Trade Agency Sandbox Portal

Comprehensive feature documentation with live
screenshots captured from the running application.

● HolyGrail v0.1.0 Next.js 15.5.20 React 19 July 2026

1. Overview

The **Trade Agency Sandbox Portal** is a full-stack advisory intelligence platform built for Canadian export trade agencies. It enables trade advisors to manage a portfolio of Small and Medium-sized Enterprises (SMEs), assess their export readiness across nine standardized pillars, and provide actionable intelligence on target markets — all within a single, premium dark-themed workspace.

NAVIGATION MODEL

ROUTE	PURPOSE
<code>/sandbox/agency</code>	Portfolio dashboard — lists all SMEs, shows aggregate readiness
<code>/sandbox/agency/report?id=<smeId></code>	Deep-dive report workspace with tabbed sections

2. Features

2.1 Portfolio Dashboard

CLIENT COMPONENT

/API/SME + /API/ASSESSMENT

FRAMER MOTION

Trade advisors manage multiple SME clients simultaneously. They need a single view to triage which companies are export-ready, which have pending assessments, and the overall health of their advisory portfolio.

CAPABILITIES

- Aggregate KPIs: **Active Clients** count and **Avg. Readiness** score with animated progress bar
- SME cards in a bento-grid layout showing company name, province, industry, target country, and readiness grade (A–F)
- "Open report" links for assessed SMEs; inline "Start questionnaire" for pending ones
- Real-time health status badge in sidebar footer (7/7 services healthy)
- Sidebar navigation with theme toggle

TECHNICAL

- Component: `AgencyPortfolioClient.tsx` — fetches `/api/sme` and `/api/assessment` in parallel
- Staggered card entrance animation using `motion` with `snappy` spring preset
- Scoring engine: `gradeColor()` and `gradeLabel()` for grade visualization

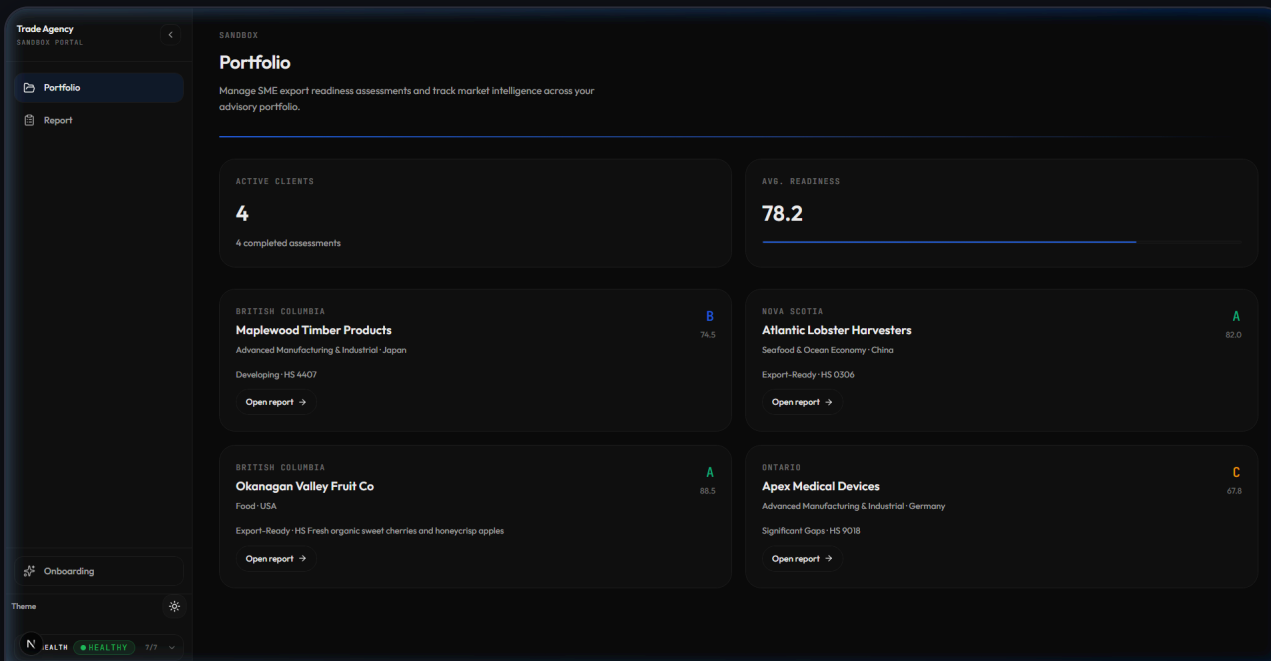


FIG 1 – Portfolio Dashboard showing 4 SMEs with readiness grades (A/A/B/C) and aggregate metrics

2.2 Readiness Score Gauge

SVG ANIMATION

USECOUNTUP HOOK

Animated SVG circular gauge displaying readiness score (0–100) and letter grade. Color-coded ring adapts to grade severity — green for A, blue for B, amber for C, red for D/F. The numeric score uses a custom `useCountUp` hook for animated decimal counting.

TECHNICAL

- Pure SVG with `motion.circle` animated `strokeDashoffset`
- Constants: `SIZE=168`, `STROKE=10`, calculated circumference for arc math
- Critically-damped spring transition preset

2.3 Sanctions & Compliance Screen

/API/SANDBOX/SCREEN

SEMA / CSL

Before proceeding with any export engagement, advisors must verify the SME isn't on any sanctions or restricted entity lists. The screen pre-fills with the company name, normalizes the query, and checks against SEMA / Consolidated Screening List data. Shows "Clear" (green shield) or "Match Found" (red alert) with matched entries and data origin audit trail.

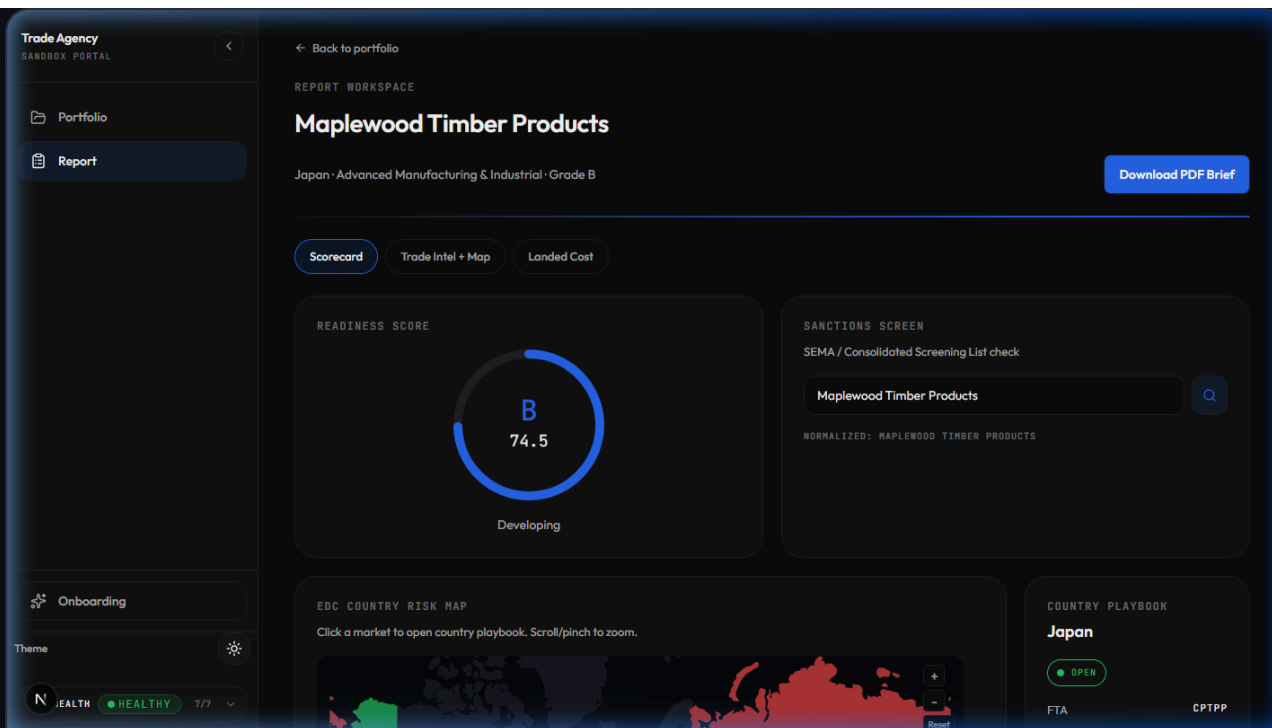


FIG 2 – Readiness Score Gauge (B / 74.5) paired with Sanctions Compliance Screen

2.4 EDC Country Risk Map & Country Playbook

@VISX/GEO

@VISX/ZOOM

TOPOJSON-CLIENT

Interactive Mercator world map colored by Export Development Canada risk tiers — **Open** (green), **Watch** (amber), **Restricted/Blocked** (red). Features an animated dashed shipping lane arc from Canada to the selected target country, with zoom/pan controls.

The **Country Playbook** side panel appears when a country is clicked, showing FTA status (e.g., CPTPP), region, default tariff, freight cost per FEU, import volume, YoY change, and sanctions clearance status.

TECHNICAL

- `@visx/geo` Mercator projection + `@visx/zoom` for pan/zoom
- World atlas topology (110m) loaded from CDN, parsed with `topojson-client`
- 28 EDC-tracked markets with ISO3 numeric mapping in `country-risk-data.ts`
- Shipping lane: SVG quadratic Bézier curve between Canada and target centroids
- Fallback: `buildMinimalWorld()` generates polygons if CDN fails

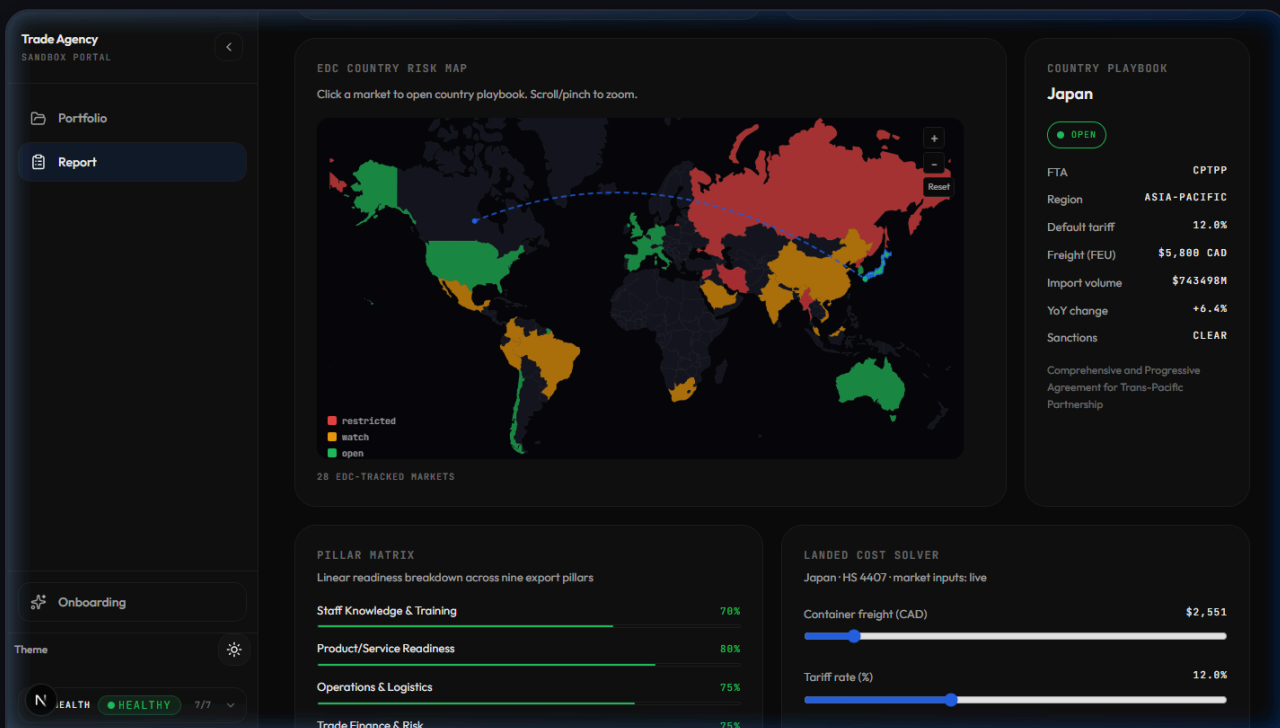


FIG 3 – EDC Country Risk Map with shipping lane to Japan + Country Playbook (FTA: CPTPP, Tariff: 12.8%)

2.5 Pillars Matrix & Critical Gap Analyzer

ANIMATED BARS

SCORING-ENGINE.TS

Displays nine export readiness pillars as animated horizontal bar charts: Staff Knowledge, Product Readiness, Operations, Trade Finance, Legal, Strategy, Cultural Readiness, Digital, and Program & Funding. Color-coded by score threshold ($\geq 70\%$ green, $\geq 50\%$ blue, $\geq 25\%$ amber, $< 25\%$ red).

The **Critical Gap Analyzer** filters pillars below a configurable threshold (default 50%) and lists them sorted by severity. When all pillars pass, a green "No critical gaps" confirmation is shown.

2.6 Landed Cost Solver

[/API/SANDBOX/FREIGHT](#)

[/API/SANDBOX/TARIFFS](#)

[/API/SANDBOX/RATES](#)

[LANDED-COST-CALCULATOR.TS](#)

Interactive cost modeling tool with three sliders — container freight (\$1,500–\$12,000 CAD), tariff rate (0–35%), and export quantity (500–50,000 units). Calculates landed cost per unit, FX-adjusted margin with currency volatility buffer, and provides a detailed cost breakdown (unit freight, tariff, broker fee, insurance).

DATA PIPELINE

- Fetches live market inputs from 3 APIs in parallel (freight, tariffs, FX rates)
- Falls back to structured country data when APIs are unavailable
- FX Buffer calculated from 30-day or 90-day currency volatility
- Data origin indicator: [live](#), [mock-fallback](#), or [structured-fallback](#)

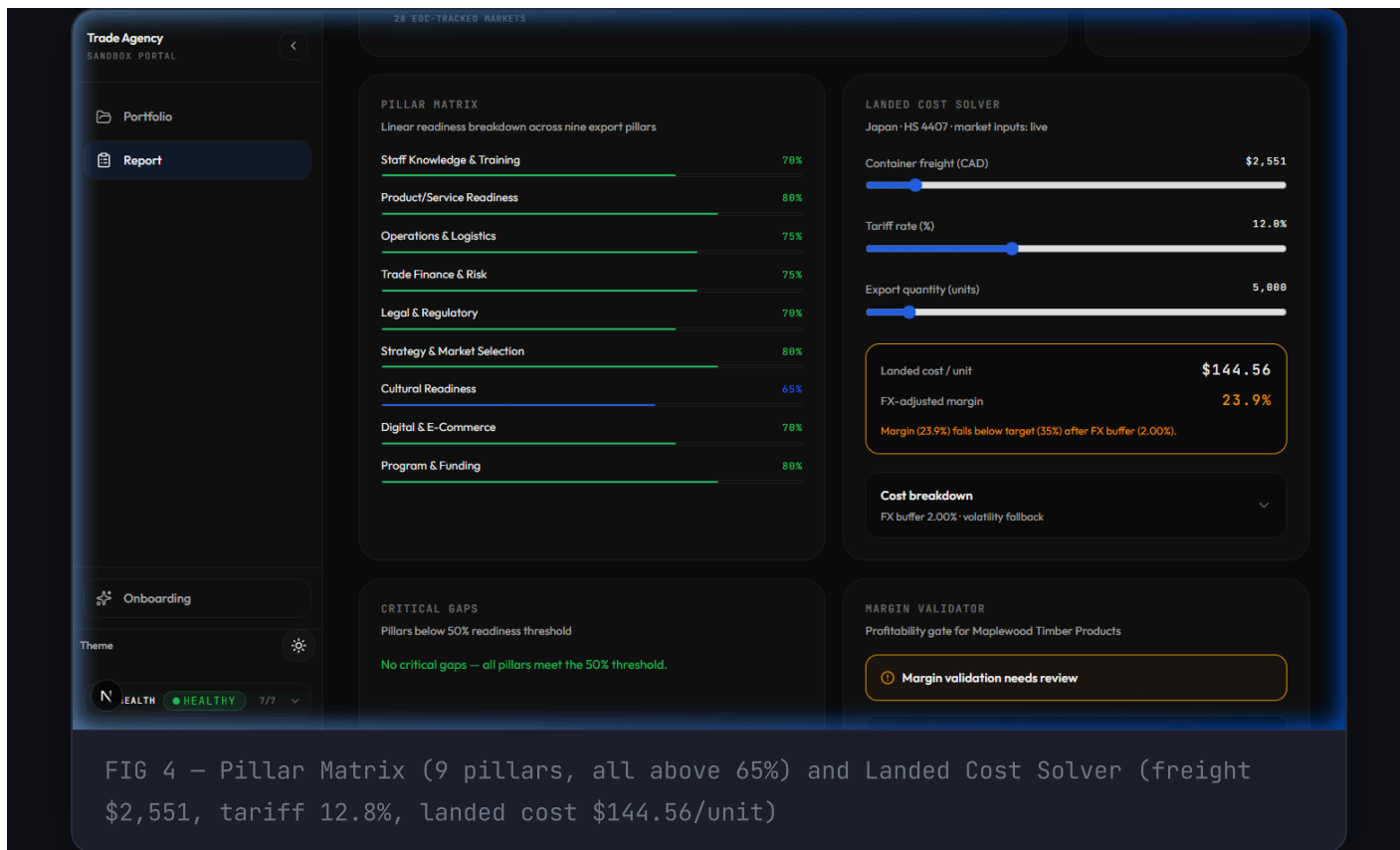


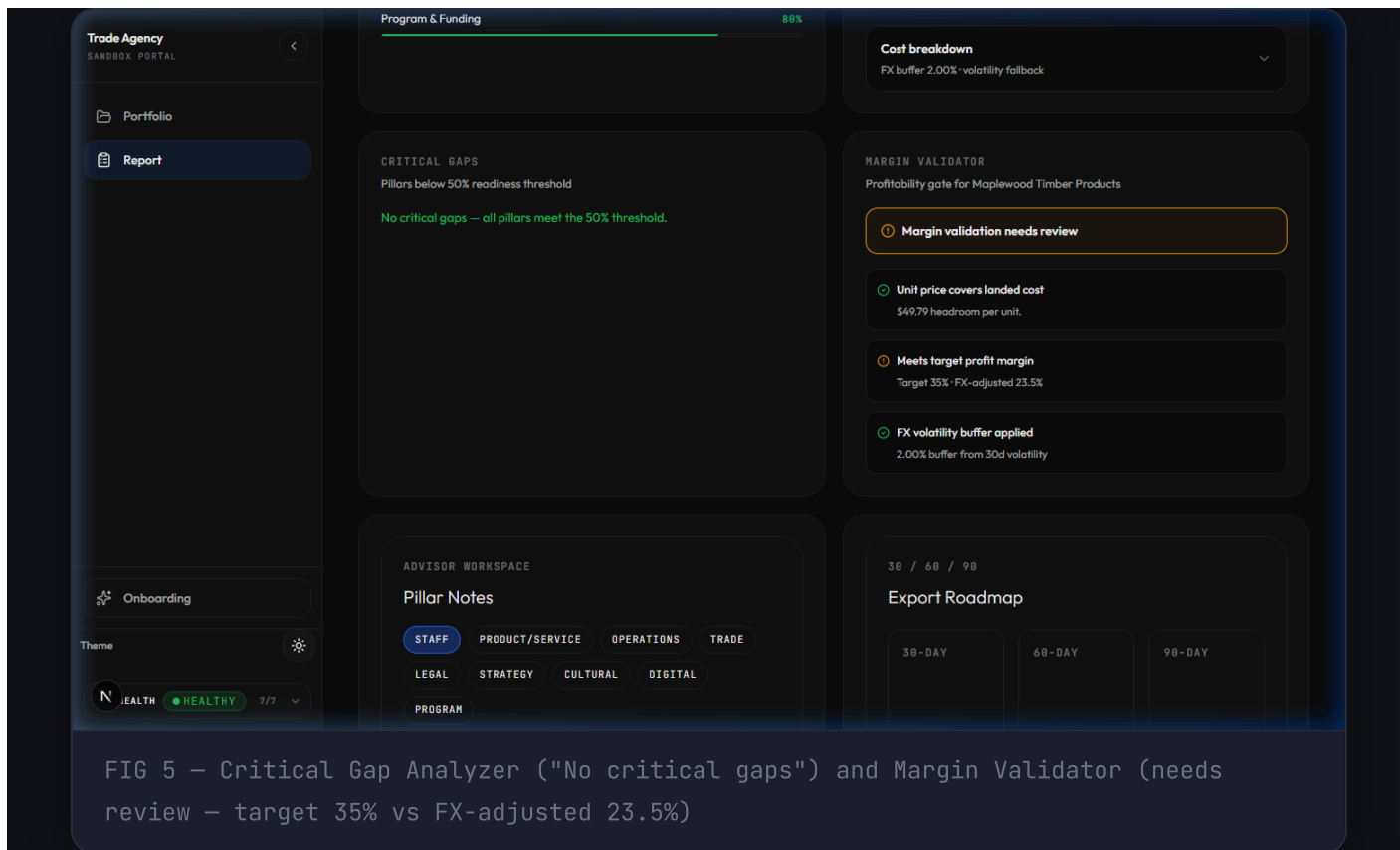
FIG 4 – Pillar Matrix (9 pillars, all above 65%) and Landed Cost Solver (freight \$2,551, tariff 12.8%, landed cost \$144.56/unit)

2.7 Margin Validator

PASS/FAIL GATE

LANDED-COST-CALCULATOR.TS

Profitability gate with three validation checks: (1) unit price covers landed cost, (2) meets target profit margin, (3) FX volatility buffer applied. Displays pass/fail status per check and an aggregate badge.



2.8 Advisor Notes Panel

[/API/ADVISOR-NOTES](#)

[AUTO-SAVE](#)

Organized by nine pillar tabs, advisors can write free-form notes per pillar. Notes are auto-saved with a 700ms debounce via [PUT /api/advisor-notes](#), with a timestamp confirmation display.

2.9 Export Roadmap Timeline (30/60/90)

[@DND-KIT/CORE](#)

[@DND-KIT/SORTABLE](#)

[/API/ROADMAP](#)

Three-column Kanban board for 30-day, 60-day, and 90-day export readiness milestones. Cards are drag-and-droppable between and within columns using [@dnd-kit](#) with pointer sensor (6px activation). Reordering is persisted via batch [PATCH /api/roadmap](#).

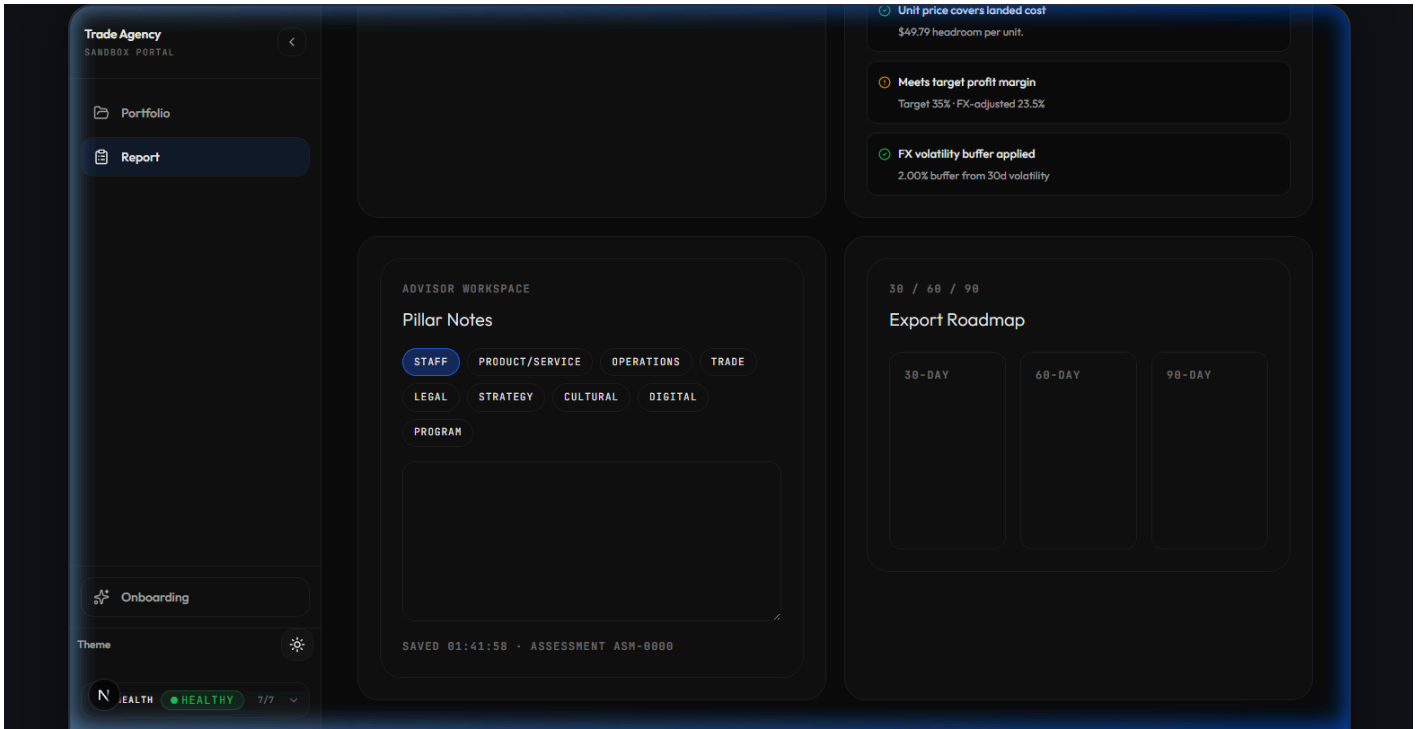


FIG 6 – Advisor Notes panel (pillar tabs with auto-save) and Export Roadmap 30/60/90 timeline

2.10 Trade Intelligence Dashboard

RECHARTS V3

/API/SANDBOX/COMTRADE

UN Comtrade-level trade data within the portal: hero metric showing Total Import Volume with YoY change, a 5-Year Import Trend line chart (2015–2024), Bilateral Market Competitors table (top 10 exporters by market share), and an Import Seasonality bar chart.

TECHNICAL

- Charts: `recharts` v3 — `LineChart`, `BarChart`, `ResponsiveContainer`
- Three parallel API calls: `type=summary`, `type=partners`, `type=trend`
- Cache layer via `comtrade-cache.ts` to reduce API calls
- Auto-formatting: millions (\$M) / billions (\$B) based on magnitude



FIG 7 – Trade Intelligence: \$1.53B total import volume (▼5.6% YoY) with 5-year trend line chart

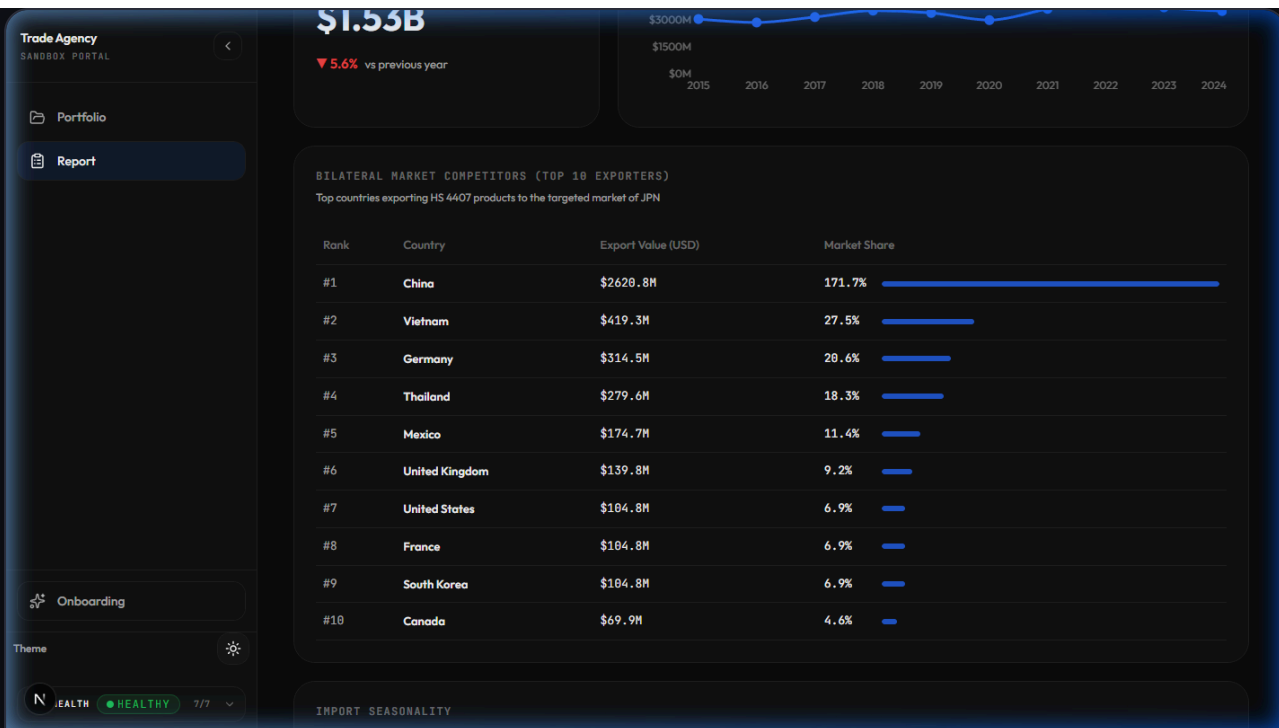


FIG 8 – Bilateral Market Competitors: Top 10 exporters of HS 4407 to Japan, ranked by market share



FIG 9 – Import Seasonality bar chart and EDC Country Risk Map in the Trade Intel tab

One-click PDF generation via the "Download PDF Brief" button. Uses server-side headless Chrome (Puppeteer) to render the report as a professional PDF. Downloads as `export-readiness-<id>.pdf` with error handling and loading states.

2.12 Landed Cost Tab (Dedicated View)

A focused tab layout pairing an expanded Landed Cost Solver (8-column span) with the Margin Validator (4-column span), followed by the full Readiness Scorecard including Score Gauge, Pillars Matrix, Critical Gaps, and AI Advisory Summary.

Maplewood Timber Products

Japan · Advanced Manufacturing & Industrial · Grade B

Download PDF Brief

Scorecard

Trade Intel + Map

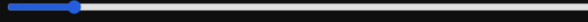
Landed Cost

LANDED COST SOLVER

Japan · HS 4407 · market inputs: live

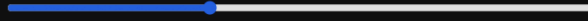
Container freight (CAD)

\$2,551



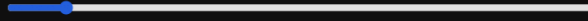
Tariff rate (%)

12.8%



Export quantity (units)

5,880



Landed cost / unit

\$144.56

FX-adjusted margin

23.9%

Margin (23.9%) falls below target (35%) after FX buffer (2.00%).

Cost breakdown

FX buffer 2.00% · volatility fallback

MARGIN VALIDATOR

Profitability gate for Maplewood Timber Products

Margin validation needs review

Unit price covers landed cost

\$49.79 headroom per unit.

Meets target profit margin

Target 35% · FX-adjusted 23.5%

FX volatility buffer applied

2.00% buffer from 30d volatility

Onboarding

Theme

⚙

N

HEALTH

HEALTHY

7/7

READINESS SCORE



Developing

PILLAR MATRIX

Linear readiness breakdown across nine export pillars

Staff Knowledge & Training	78%
Product/Service Readiness	80%
Operations & Logistics	75%
Trade Finance & Risk	75%
Legal & Regulatory	78%
Strategy & Market Selection	80%
Cultural Readiness	65%
Digital & E-Commerce	70%
Program & Funding	80%

CRITICAL GAPS

Pillars below 50% readiness threshold

No critical gaps — all pillars meet the 50% threshold.

AI ADVISORY SUMMARY

Maplewood Timber Products is moderately ready for Japan (Grade B). Core readiness is stable, but cultural adaptation and local standards compliance (JAS certification) require attention.

FIG 10 — Dedicated Landed Cost tab with expanded solver, margin validator, scorecard, and AI advisory summary

3. Tech Stack Summary

Frontend

LAYER	TECHNOLOGY	VERSION
Framework	Next.js (App Router)	15.5.20
Runtime	React	19.1.0
Language	TypeScript	5.x
Bundler	Turbopack	built-in
Styling	Tailwind CSS + custom CSS variables	3.4.17
Typography	Outfit + JetBrains Mono (Google Fonts)	—
Icons	Lucide React	1.24.0

Visualization & Interaction

PURPOSE	LIBRARY	USAGE
Charts	<code>recharts</code> v3.9	Line charts, bar charts, tooltips
Geo Map	<code>@visx/geo</code> v4	Mercator world projection
Map Zoom	<code>@visx/zoom</code> v4	Pan/zoom/reset controls
Topology	<code>topojson-client</code> v3	TopoJSON → GeoJSON parsing
Animation	<code>motion</code> v12.42	Spring transitions, layout animations
Drag & Drop	<code>@dnd-kit</code> v6+v10	Roadmap kanban board

Backend & Data

LAYER	TECHNOLOGY	PURPOSE
Database	Supabase	SMEs, assessments, notes persistence
AI	Google Generative AI (Gemini)	AI advisory summaries
PDF	Puppeteer v25	Server-side PDF rendering
Validation	Zod v4	Schema validation
Fallback DB	CSV-DB / In-Memory Store	Offline data layer

External Data Sources

SOURCE	DATA
UN Comtrade	Import volumes, bilateral partners, multi-year trends
Tariff DB	HS code tariff rates by country
Freight Rates	Container shipping rates by route
FX Rates	Exchange rates + 30d/90d volatility
SEMA / CSL	Sanctions entity screening
EDC Risk	28 country risk classifications